

What the optimal relaxation parameter really buys — classical iterative solvers on the 2D Poisson problem, predicted versus measured

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Abstract

We compare measured iteration counts to fixed relative-residual tolerance with textbook asymptotic predictions for Jacobi, Gauss-Seidel, and SOR with optimal ω^* on the 5-point discretization of $-\Delta u = 1$ on the unit square with homogeneous Dirichlet conditions for $n=16, 32, 64$. Jacobi and Gauss-Seidel match predictions closely; SOR with ω^* is dramatically faster but needs $\sim 30\%$ more iterations than the spectral-radius prediction, which we leave as an open question.

1 Introduction

Classical stationary iterative methods—Jacobi, Gauss–Seidel, and successive over-relaxation (SOR)—are typically the first examples encountered when the numerical solution of the Poisson equation is introduced. Their convergence theory is deceptively clean. For the five-point discretization of $-\Delta u = 1$ on the unit square with homogeneous Dirichlet conditions, the iteration matrices have spectral radii $\rho_J = \cos(\pi h)$, $\rho_{GS} = \rho_J^2$, and, for the classical optimal relaxation parameter $\omega^* = 2/(1 + \sin(\pi h))$, $\rho_{SOR} = \omega^* - 1$. These quantities give the asymptotic per-iteration contraction factor of the error. What a computation actually reports, however, is the number of iterations needed to drive some stopping criterion below a tolerance, and the asymptotic factor is only an indirect predictor of that count: it must first be converted into an estimated iteration count under the assumption that the asymptotic regime has set in. Textbook discussions rarely perform this conversion explicitly, and students are left with little guidance about how much trust to place in the predicted counts at moderate grid sizes.

This short note is intended as a reproducible teaching reference that closes that gap. We compare measured iteration counts with spectral-radius predictions for the model problem $-\Delta u = 1$ on the unit square, using the standard five-point stencil on $n \times n$ interior grids with $h = 1/(n+1)$ and the zero initial guess. The stopping rule is the relative residual criterion $\|r_k\|_2/\|r_0\|_2 < 10^{-8}$. We consider Jacobi, Gauss–Seidel, and SOR with the optimal parameter ω^* , for $n = 16, 32$, and 64 .

The measurements reproduce the textbook scaling: the Jacobi and Gauss–Seidel iteration counts grow roughly like n^2 , while the SOR count grows like n ; doubling n multiplies the first two counts by about 3.8–3.9 and the SOR count by about 2. Jacobi and Gauss–Seidel also match the spectral-radius prediction to within a few percent, and Gauss–Seidel requires almost exactly half as many iterations as Jacobi, as $\rho_{GS} = \rho_J^2$ suggests. SOR with ω^* is dramatically faster than both methods, but its measured count exceeds the prediction derived from ρ_{SOR} by roughly 30–37% on these grids. We do not claim to have explained this gap; the most plausible hypothesis, that

a transient associated with the non-normal SOR iteration matrix delays the asymptotic regime, remains untested here.

The remainder of the paper is organized as follows. Section 2 defines the model problem, discretization, and the three iterative methods. Section 3 records the spectral radii and the predicted iteration counts. Section 4 describes the numerical experiment, gives the measured iteration counts, and compares them with the predictions. Section 5 discusses the SOR gap and lists the questions we leave open.

2 Related Work

Successive over-relaxation (SOR) was devised simultaneously by David M. Young Jr. and Stanley P. Frankel in 1950, specifically to make iterative linear solvers suitable for automatic computation on digital computers; earlier over-relaxation methods, such as those of Richardson and Southwell, had been designed for human calculators and required expert judgment [1]. According to the Wikipedia article, these historical aspects are discussed in Young’s thesis [1]; the thesis itself was not consulted for the present work. The method is a variant of Gauss–Seidel in which the update is weighted by a relaxation factor ω , with $\omega > 1$ for over-relaxation [1]. For a symmetric positive-definite matrix, Ostrowski’s theorem guarantees convergence of SOR for $0 < \omega < 2$ [1]. Under the consistent-ordering assumptions used in the convergence theory for the model problem, the Gauss–Seidel iteration matrix has spectral radius $\rho_{\text{GS}} = \mu^2$, and the optimal relaxation parameter is

$$\omega_{\text{opt}} = \frac{2}{1 + \sqrt{1 - \mu^2}},$$

where μ is the spectral radius of the Jacobi iteration matrix [1]. For the five-point discretization of $-\Delta u = 1$ on the unit square, $\mu = \cos(\pi h)$, and this formula reduces to $\omega^* = 2/(1 + \sin(\pi h))$.

The present overview is deliberately narrow. Of the standard references often cited in this area, only the Wikipedia article on SOR has been read in full; Young’s thesis, Varga’s *Matrix Iterative Analysis*, Golub and Van Loan’s text, and LeVeque’s finite-difference text remain unchecked. The numerical results in this paper are therefore compared with the textbook formulas listed above rather than with a systematic reading of the original literature. The experiments reported below test how well these textbook predictions describe measured iteration counts at moderate grid sizes.

3 Model Problem and Discretization

We consider the two-dimensional Poisson equation with constant source term,

$$-\Delta u = 1 \quad \text{in } \Omega = (0, 1)^2, \quad u = 0 \quad \text{on } \partial\Omega. \quad (1)$$

Let n be the number of interior grid points in each direction and set $h = 1/(n + 1)$. The grid points are $(x_i, y_j) = (ih, jh)$, $i, j = 0, \dots, n + 1$; the unknowns are the approximate values $u_{i,j} \approx u(x_i, y_j)$ at the interior points with $1 \leq i, j \leq n$. Using the standard five-point stencil for the Laplacian and incorporating the homogeneous Dirichlet boundary conditions by setting all boundary values to zero gives the discrete equations

$$\frac{4u_{i,j} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1}}{h^2} = 1, \quad 1 \leq i, j \leq n, \quad (2)$$

with $u_{0,j} = u_{n+1,j} = u_{i,0} = u_{i,n+1} = 0$. In matrix form this is $A_h u_h = \mathbf{1}$, where A_h is the $n^2 \times n^2$ discrete Dirichlet Laplacian and $\mathbf{1} \in \mathbb{R}^{n^2}$ is the vector of all ones.

The computations reported in this paper use only the three mesh sizes $n = 16, 32, 64$, corresponding to $h = 1/17, 1/33, 1/65$ and to $n^2 = 256, 1024, 4096$ unknowns.

4 Methods

We solve the two-dimensional Poisson problem

$$-\Delta u = 1 \quad \text{in } (0, 1)^2, \quad u = 0 \quad \text{on } \partial(0, 1)^2,$$

with the standard five-point stencil on an $n \times n$ interior grid. The mesh width is $h = 1/(n + 1)$, and the resulting linear system is $A_h u_h = b_h$. We compare three stationary iterations: Jacobi, Gauss–Seidel, and successive over-relaxation (SOR) with the classical optimal relaxation parameter

$$\omega^* = \frac{2}{1 + \sin(\pi h)}.$$

For this model problem the associated spectral radii are

$$\rho_J = \cos(\pi h), \quad \rho_{GS} = \rho_J^2, \quad \rho_{SOR} = \omega^* - 1,$$

and the standard asymptotic estimate for the iteration count at tolerance ϵ is

$$K_{\text{pred}} = \left\lceil \frac{\ln \epsilon}{\ln \rho} \right\rceil.$$

We used $\epsilon = 10^{-8}$, grids $n = 16, 32, 64$, and the corresponding ω^* values in Table 1. In all runs the initial guess was $u_0 = 0$, and the stopping criterion was

$$\|r_k\|_2 / \|r_0\|_2 < 10^{-8},$$

where $r_k = b_h - A_h u_k$ is the residual after k iterations.

All iterations were implemented with red-black ordering to permit vectorized sweeps in NumPy. Under the standard consistently-ordered theory for the five-point Laplacian, this ordering is expected to have the same convergence rate as lexicographic ordering; we do not prove that here, and the measured iteration counts below are for the red-black implementation.

n	h	ω^*
16	1/17	1.6895
32	1/33	1.8264
64	1/65	1.9078

Table 1: Relaxation parameters used in the SOR experiments.

5 Theoretical Predictions

Classical spectral-radius analysis for the five-point Laplacian on a uniform grid with homogeneous Dirichlet boundary conditions gives the following predictions. Let $h = 1/(n + 1)$ and let ρ_J, ρ_{GS} ,

and ρ_{SOR} denote the spectral radii of the Jacobi, Gauss–Seidel, and SOR iteration matrices. For this consistently ordered model problem,

$$\rho_J = \cos(\pi h), \quad \rho_{GS} = \rho_J^2, \quad \omega^* = \frac{2}{1 + \sin(\pi h)}, \quad \rho_{SOR} = \omega^* - 1.$$

Using the spectral radius as the asymptotic per-iteration reduction factor, the predicted iteration count for a relative residual tolerance $\epsilon = 10^{-8}$ is

$$N_{\text{pred}}(\rho) = \left\lceil \frac{\ln(10^{-8})}{\ln \rho} \right\rceil.$$

For $n = 16, 32, 64$, the resulting predicted counts are shown in Table 2. These predictions exhibit the

	$n = 16$	$n = 32$	$n = 64$
ω^*	1.6895	1.8264	1.9078
Jacobi	1073	4059	15765
Gauss–Seidel	537	2030	7883
SOR(ω^*)	50	97	191

Table 2: Predicted iteration counts from the spectral-radius formula with $\epsilon = 10^{-8}$, rounded up to the nearest integer.

standard asymptotic scaling: doubling n multiplies the Jacobi and Gauss–Seidel counts by roughly a factor of four, while the optimal SOR count only doubles, corresponding to $O(n^2)$ iterations for Jacobi and Gauss–Seidel and $O(n)$ iterations for SOR(ω^*).

6 Experimental Setup

All experiments solve the two-dimensional model problem $-\Delta u = 1$ on the unit square with homogeneous Dirichlet boundary conditions, discretized by the standard five-point stencil on an $n \times n$ grid of interior points. The mesh size is $h = 1/(n+1)$, and the number of unknowns is n^2 ; we use $n = 16, 32, 64$. The right-hand side is the constant function $f = 1$.

Three stationary iterative methods are compared: Jacobi, Gauss–Seidel, and SOR with the classical optimal relaxation parameter $\omega_* = 2/(1 + \sin(\pi h))$. The implementation is written in pure numpy, and the Gauss–Seidel and SOR sweeps use red-black ordering so that the updates can be vectorized; for the five-point Laplacian this ordering has the same asymptotic convergence rate as lexicographic ordering. All runs start from the zero initial guess $u_0 = 0$ and use the same right-hand side.

Every run is stopped as soon as the relative residual satisfies

$$\frac{\|r_k\|_2}{\|r_0\|_2} < 10^{-8},$$

where $r_k = b - Au_k$ and A is the standard five-point discrete Laplacian. Since $u_0 = 0$ and $f = 1$, the initial residual is simply $r_0 = b$. No other stopping criterion was used.

Only the grids $n = 16, 32, 64$ are considered. The next grid, $n = 128$, was not run: the pure numpy implementation was too slow to complete it in the available time. Accordingly, no claim in this paper is made beyond $n = 64$.

7 Results

8 Discussion

For Jacobi and Gauss-Seidel, the textbook spectral-radius prediction is in excellent quantitative agreement with the measured iteration counts. Using the standard five-point stencil on an $n \times n$ interior grid, the relevant radii are

$$\rho_J = \cos \pi h, \quad \rho_{GS} = \rho_J^2, \quad h = 1/(n+1),$$

and the standard spectral-radius estimate for the iteration count is

$$k_{\text{pred}} \approx \frac{\ln(10^{-8})}{\ln \rho}.$$

For $n = 16, 32, 64$, the measured Jacobi counts are 1064, 4020, and 15599, against predictions 1073, 4059, and 15765; the corresponding ratios are 0.99, 0.99, and 0.99. Gauss-Seidel is equally well described: measured 543, 2048, 7948 versus predicted 537, 2030, 7883, with ratios 1.01, 1.01, and 1.01. The relation $\rho_{GS} = \rho_J^2$ also appears directly in the counts: Gauss-Seidel requires almost exactly half as many iterations as Jacobi on every grid tested. Doubling n multiplies both counts by roughly 3.8–3.9, consistent with the expected $O(n^2)$ growth.

SOR with the classical optimal parameter

$$\omega^* = \frac{2}{1 + \sin \pi h}, \quad \rho_{SOR} = \omega^* - 1,$$

behaves very differently. The measured counts, 65, 129, and 261 for $n = 16, 32, 64$, are dramatically smaller than the Gauss-Seidel counts and grow by about a factor of 2 when n doubles, matching the expected $O(n)$ rate. However, they are systematically larger than the spectral-radius prediction: the predicted counts are 50, 97, and 191, giving measured-to-predicted ratios 1.30, 1.33, and 1.37. The SOR gap is therefore not a small correction: on these grids the method needs roughly 30–37% more iterations than the spectral-radius estimate predicts.

We leave this SOR gap as an open question. The cause is not established by the present experiments, and we make no claim about the mechanism.

9 Limitations and Future Work

This study is deliberately narrow. All reported measurements are for the 5-point discretization of $-\Delta u = 1$ on the unit square with homogeneous Dirichlet conditions, the zero initial guess, the relative-residual stopping criterion $\|r_k\|_2/\|r_0\|_2 < 10^{-8}$, and grid sizes $n = 16, 32, 64$. The implementation was written in pure NumPy; the SOR runs used red-black ordering for vectorization, and we did not compare red-black with lexicographic ordering. Consequently, the quantitative conclusions should not be extrapolated to other right-hand sides, initial guesses, tolerances, or much larger grids without additional experiments. In particular, $n = 128$ was not run because the pure-NumPy implementation was too slow.

Within this scope, the main empirical findings are clear: Jacobi and Gauss-Seidel iteration counts grow like n^2 and match spectral-radius predictions to within a few percent (with Gauss-Seidel about half of Jacobi), while SOR with ω^* grows like n but requires roughly 30–37% more iterations than the prediction based on $\rho_{SOR} = \omega^* - 1$. The observed excess ratios were 1.30, 1.33, and 1.37 for $n = 16, 32, 64$; with only three grid sizes we do not infer a trend. We do not claim an explanation

for this SOR gap, and we leave it as an open question. The non-normality of the SOR iteration matrix suggests a transient effect as a possible cause, but this is a conjecture, not a result, and we did not test it.

Future work could address the main limitations in four ways. First, run $n = 128$ (with a compiled or sparse implementation) to check whether the SOR gap persists or grows. Second, sweep ω around ω^* to characterize the sensitivity of the iteration count to the relaxation parameter; we did not perform such a sweep. Third, record residual histories and explicitly test the transient hypothesis, for example by checking for a nonmonotone residual hump in the SOR iterations. Fourth, compare against conjugate gradient and multigrid, which were noted as natural out-of-scope comparisons. We emphasize that none of these four experiments was performed here.

References

- [1] Wikipedia contributors. *Successive over-relaxation*. Wikipedia, The Free Encyclopedia. URL: https://en.wikipedia.org/wiki/Successive_over-relaxation.